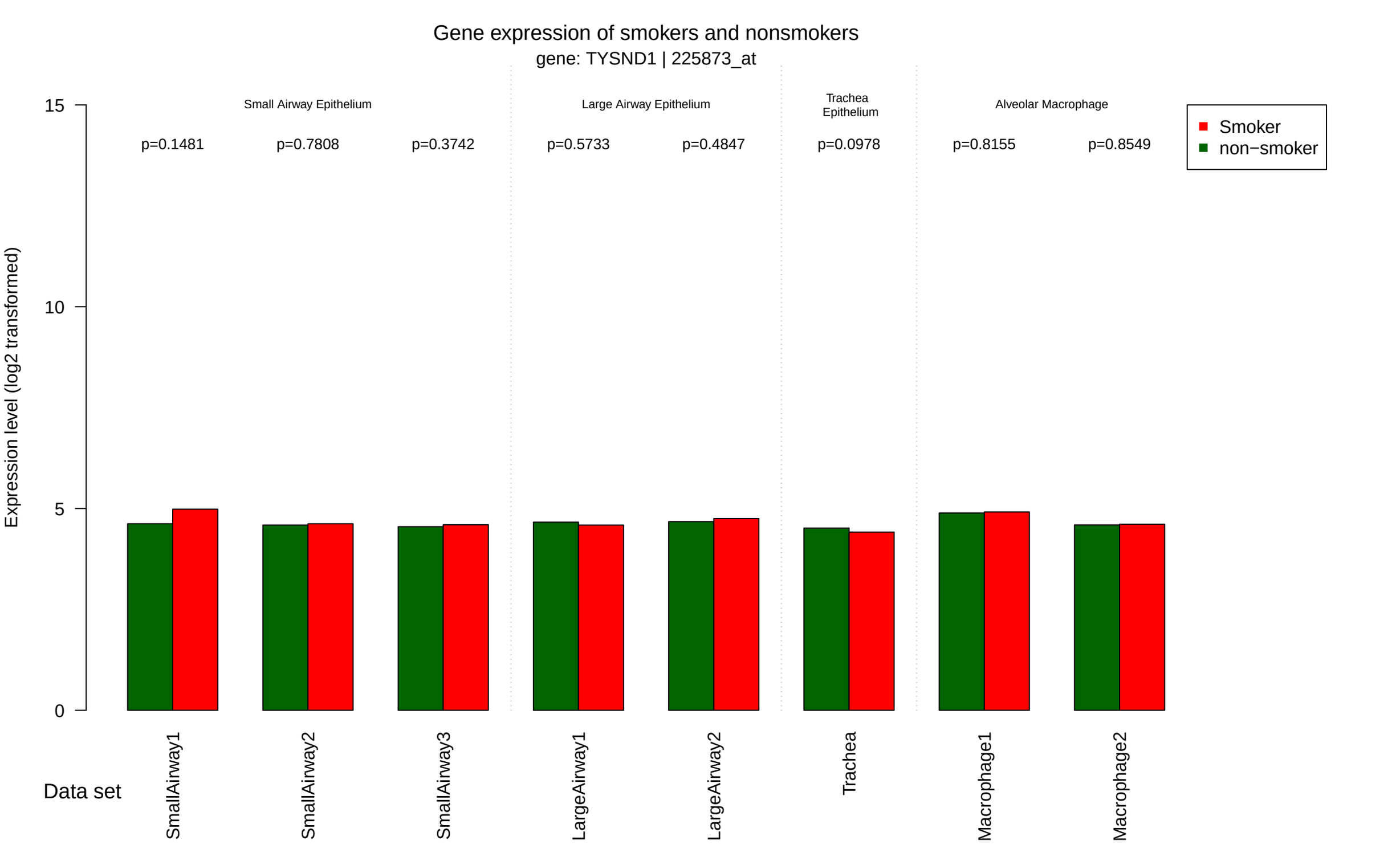
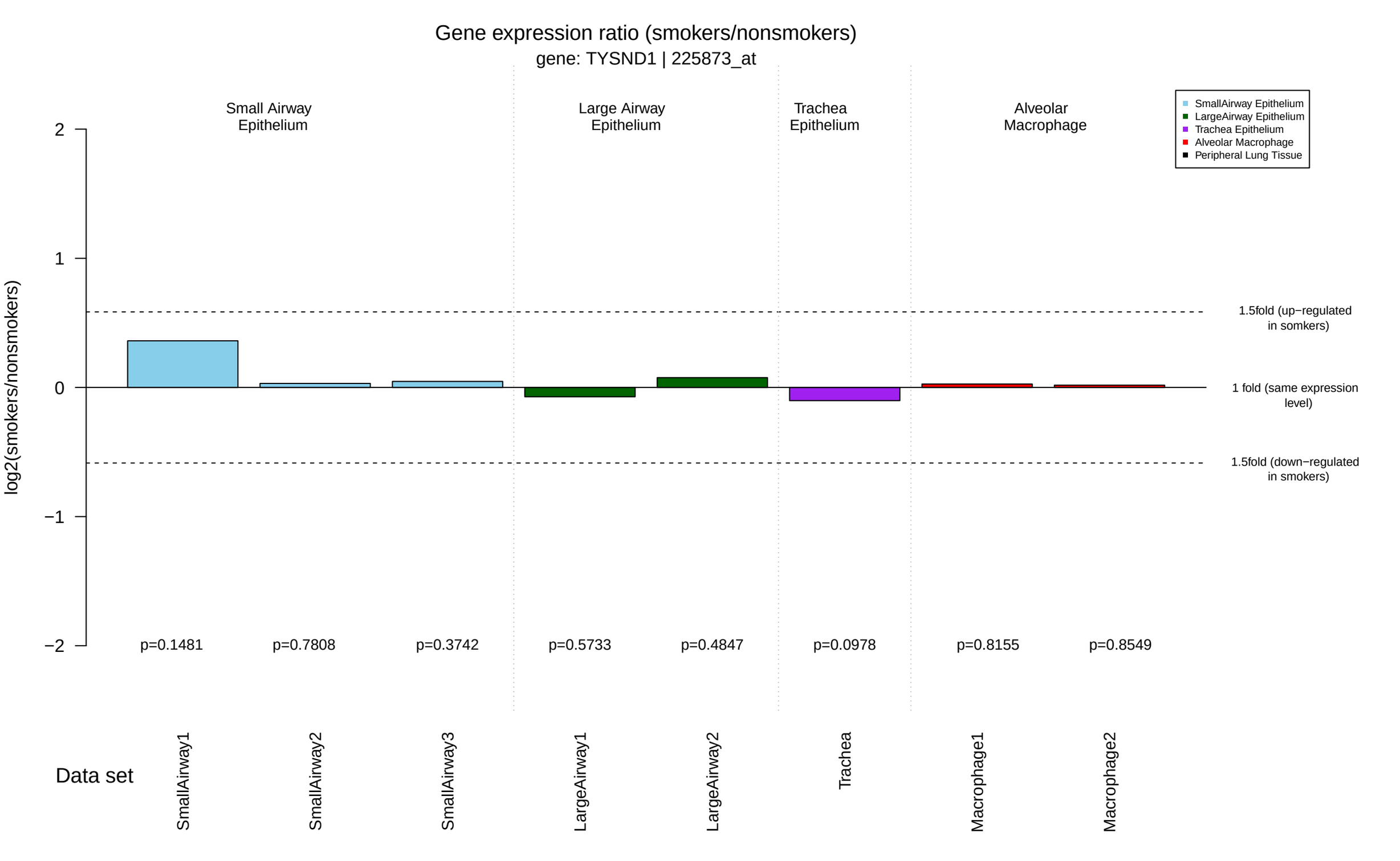
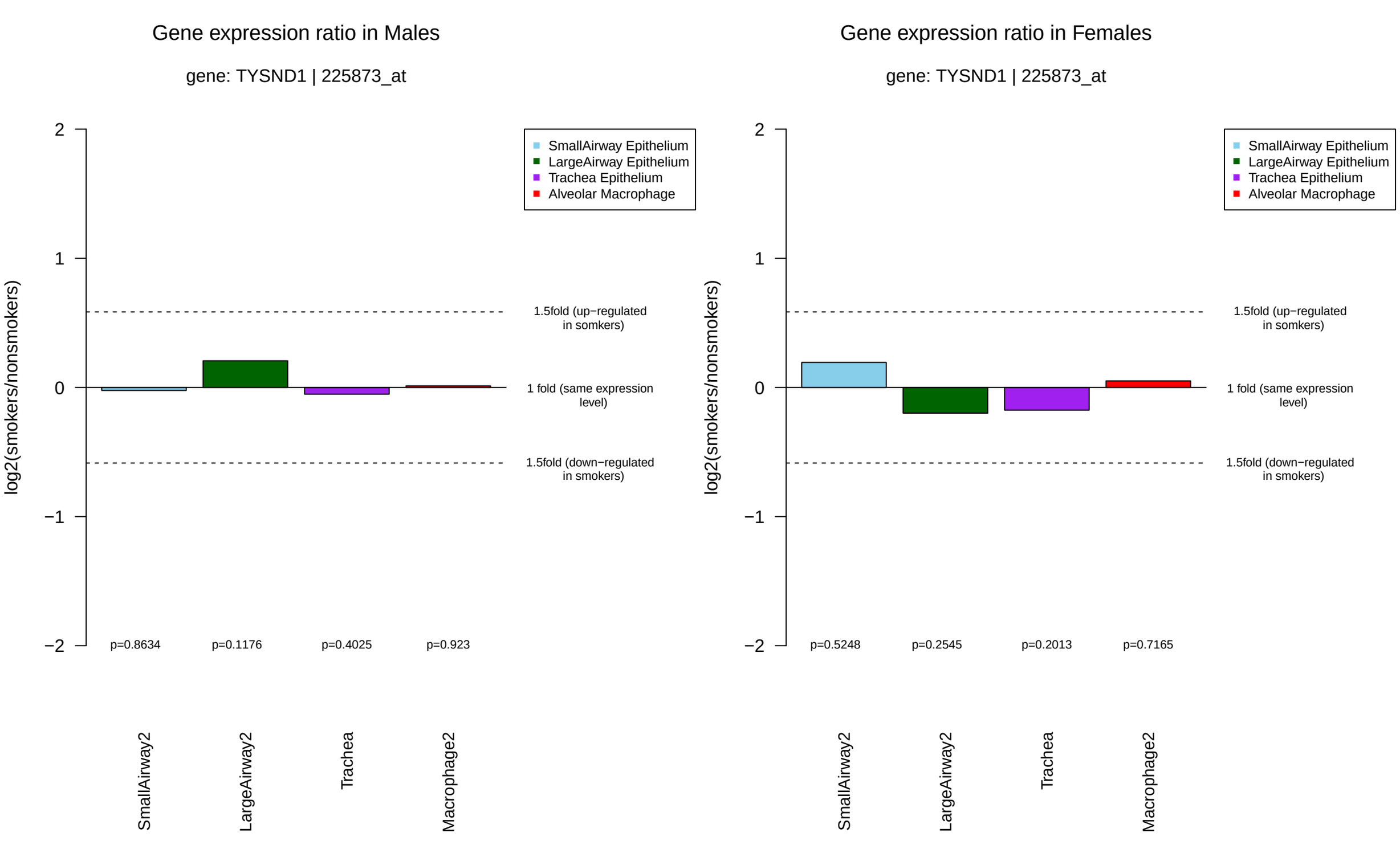
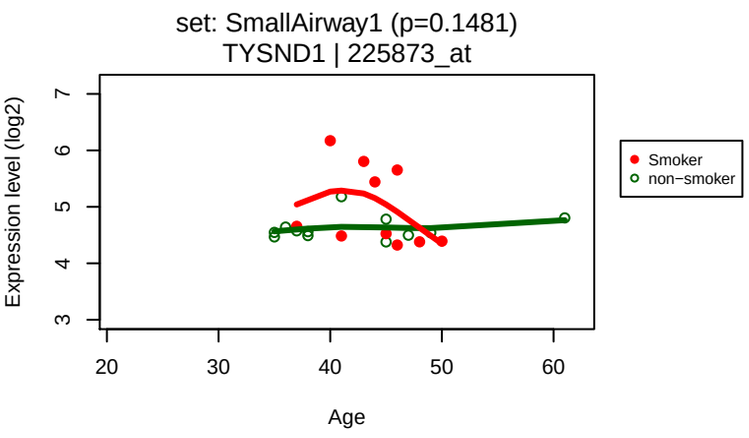
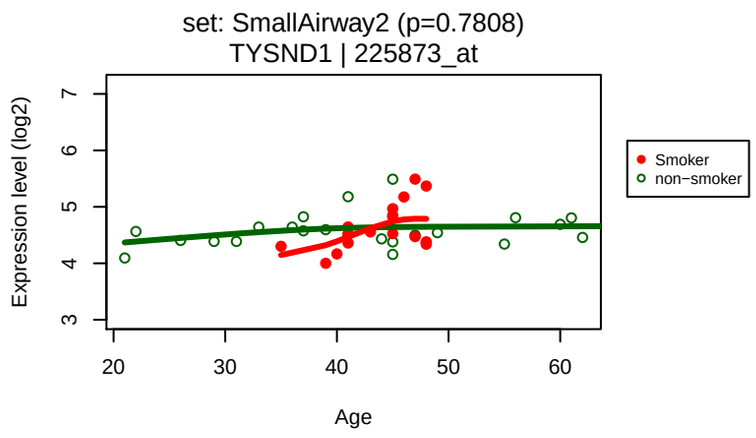
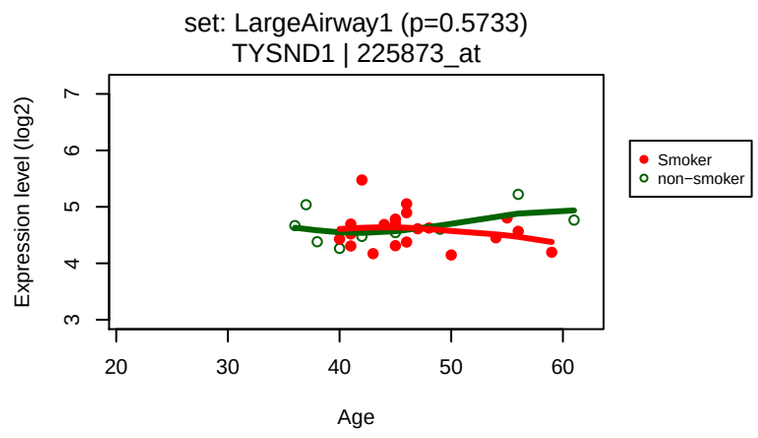

Gene expression of smokers and nonsmokers
gene: TYSND1 | 225873_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1481
p=0.7808
p=0.3742
p=0.5733
p=0.4847
p=0.0978
p=0.8155
p=0.8549
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

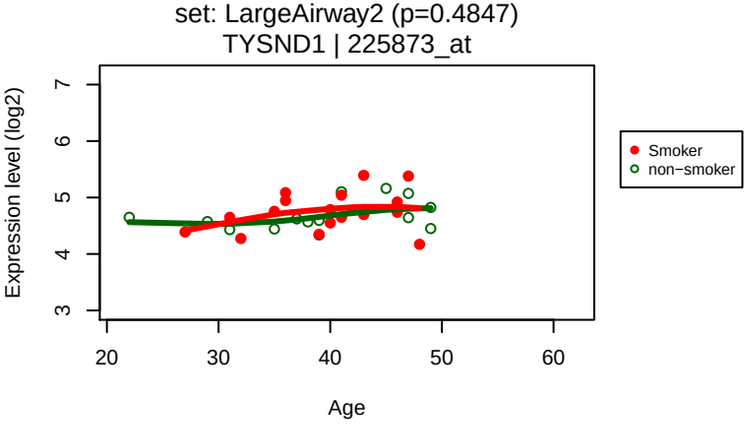

Gene expression ratio (smokers/nonsmokers)
gene: TYSND1 | 225873_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1481
p=0.7808
p=0.3742
p=0.5733
p=0.4847
p=0.0978
p=0.8155
p=0.8549
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

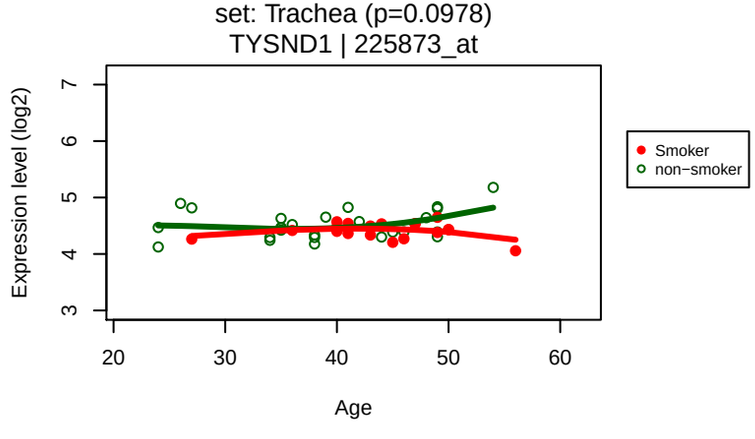

Gene expression ratio in Males
gene: TYSND1 | 225873_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8634
p=0.1176
p=0.4025
p=0.923
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TYSND1 | 225873_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5248
p=0.2545
p=0.2013
p=0.7165
SmallAirway2
LargeAirway2
Trachea
Macrophage2

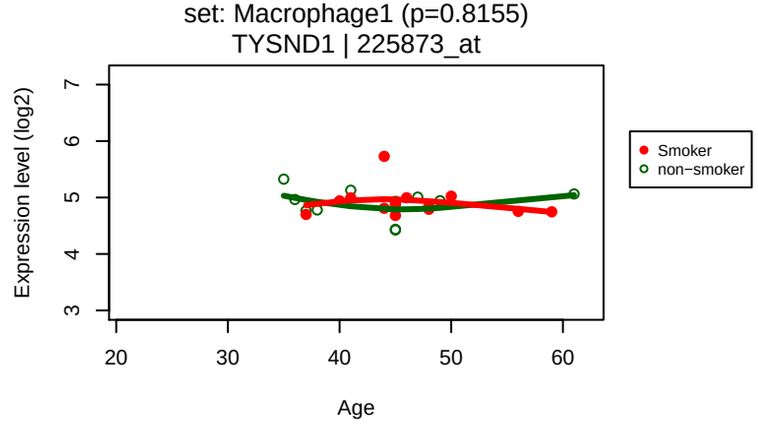

set: SmallAirway1 (p=0.1481)
TYSND1 | 225873_at
Expression level (log2)
Age
Smoker
non−smoker

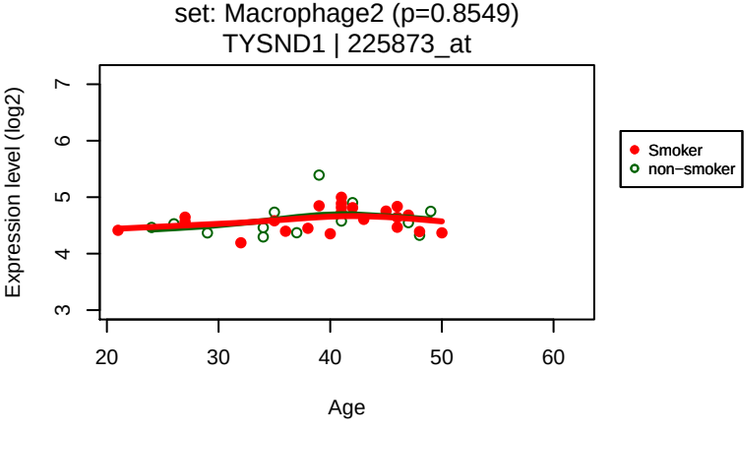

set: SmallAirway2 (p=0.7808)
TYSND1 | 225873_at
Expression level (log2)
Age
Smoker
non−smoker

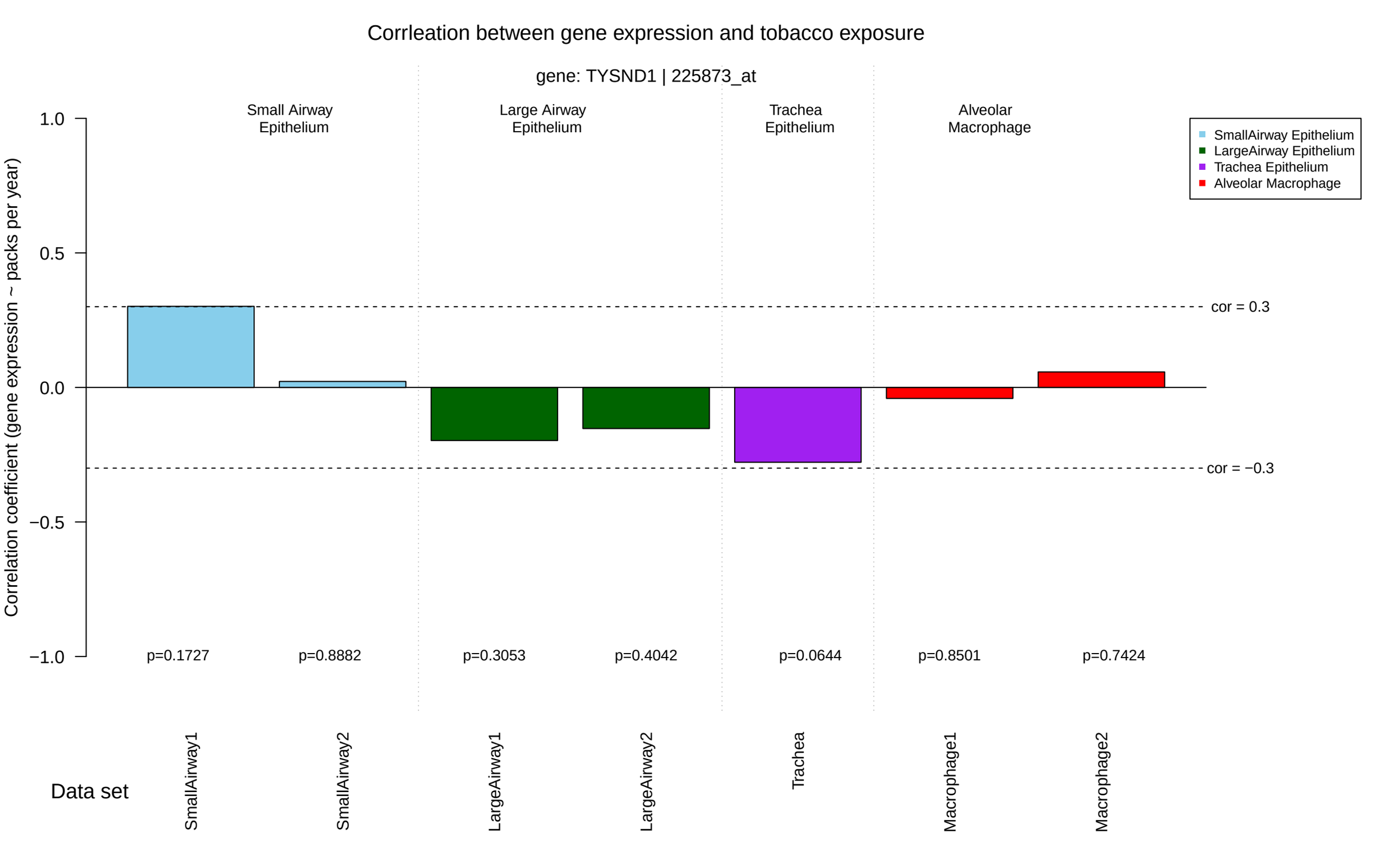

set: LargeAirway1 (p=0.5733)
TYSND1 | 225873_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4847)
TYSND1 | 225873_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0978)
TYSND1 | 225873_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8155)
TYSND1 | 225873_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8549)
TYSND1 | 225873_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TYSND1 | 225873_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1727
p=0.8882
p=0.3053
p=0.4042
p=0.0644
p=0.8501
p=0.7424
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2